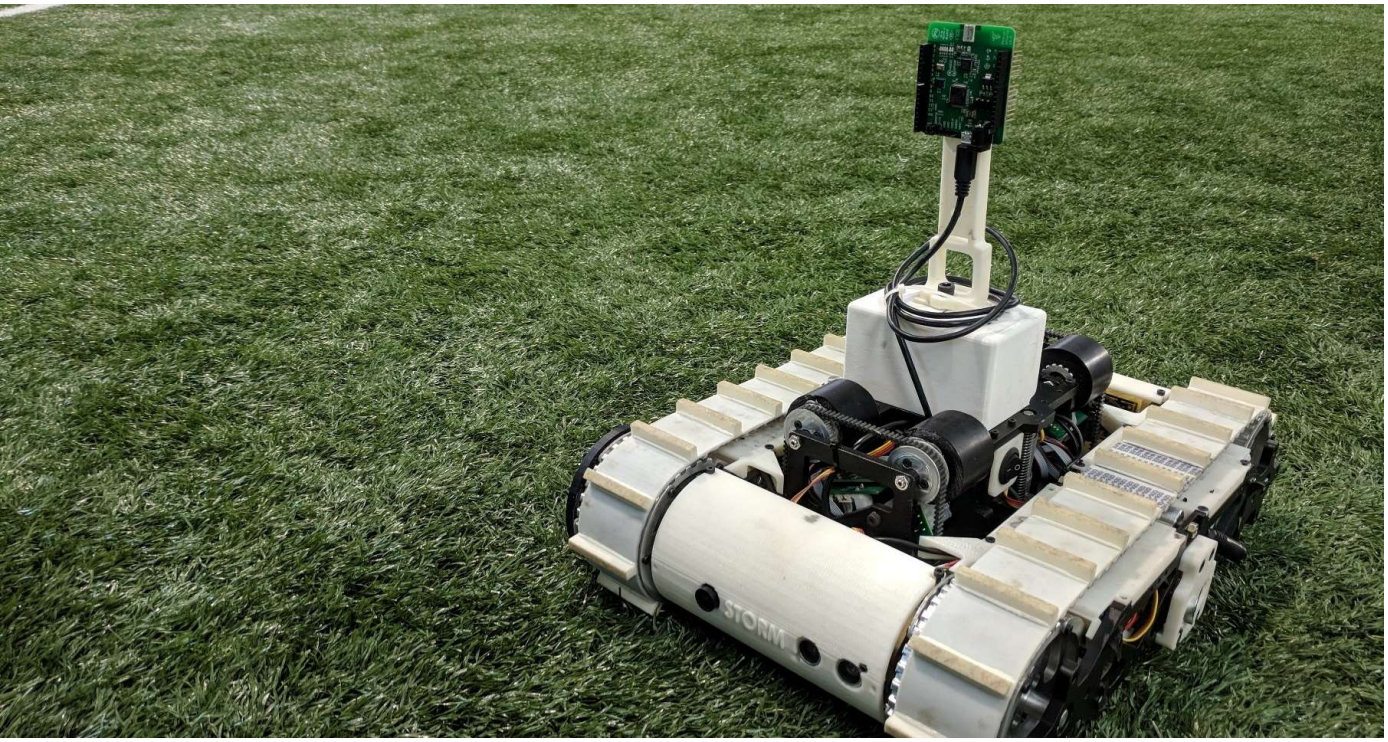


INTRODUCTION TO FIELD AND SERVICE ROBOTICS

Instructor: Bijo Sebastian

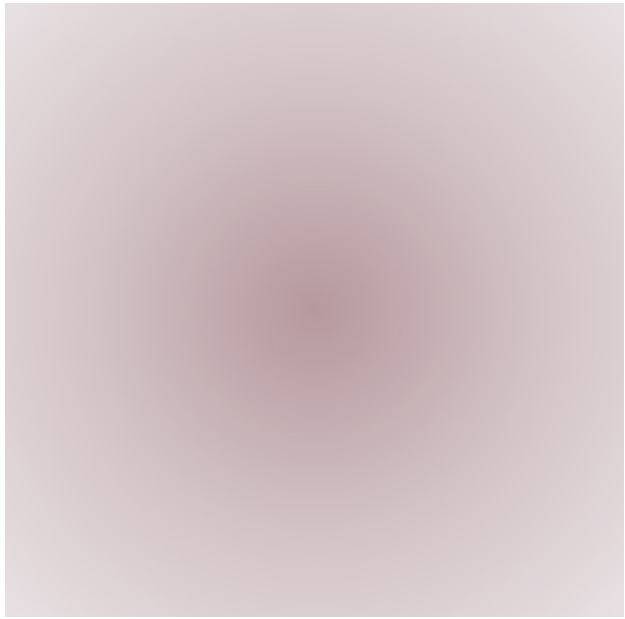
ED 5315: INTRO TO FIELD & SERVICE ROBOTICS

Instructor:

Bijo Sebastian
Assistant Professor,
Department of Engineering Design
Email: bijo.sebastian@iitm.ac.in
Lab website: https://asl-iitm.github.io/

Teaching Assistants:

Devika: devika.ag007@gmail.com
Gowry: gowrynandaps@gmail.com



ELIGIBILITY

Who can take this course:

- IDDD Robotics students (Core basket course)
- M. Tech Robotics students (Core basket course)
- Other B. Tech students → As elective; subject to course limit
- MS PhD Students → Subject to approval from your guide
- Staff (IIT-M and RP) → Subject to approval through SSP portal

IDDD ROBOTICS

Inter Disciplinary Dual Degree Program in Robotics (from Jan 2026)

Three core courses, one seminar, one lab + four electives

Covering the mechanical, electrical, computer science fundamentals, and applications.

- ID 6040: Introduction to Robotics
- ED 6007: Mechanics of Robots
- ED 5315: Introduction to Field and Service Robotics
- ED 5215: Introduction to Motion Planning
- ID 6071: Robot control
- ID 5201: Robotics Seminar
- ID 6100: Robotics Laboratory
- + Four electives from the approved list

2 out of 4

For more details refer:

https://ed.iitm.ac.in/site_resources/programs/docs/IDDD_Robotics_2026-2.pdf

M. TECH. ROBOTICS

New program starting from August 2026

Four core courses, one seminar, one lab + six electives

Covering the mechanical, electrical, computer science fundamentals, and applications.

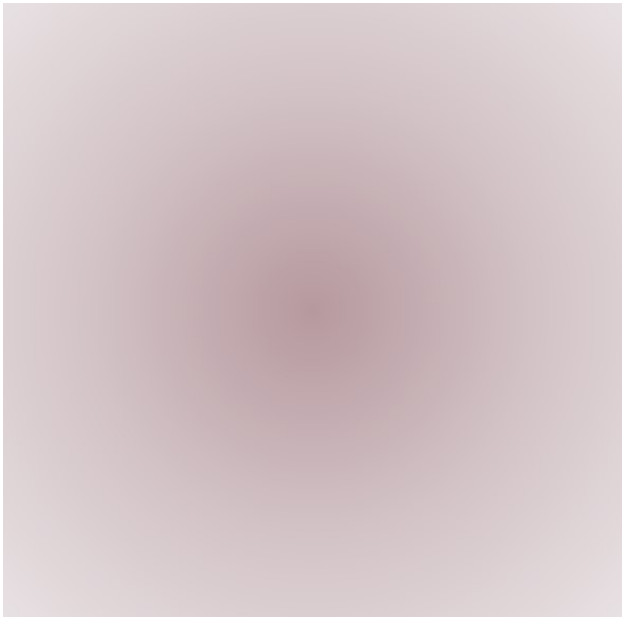
- ID 6040: Introduction to Robotics
- ID 6071: Robot control
- ED 6007: Mechanics of Robots
- ED 5315: Introduction to Field and Service Robotics
- ED 5215: Introduction to Motion Planning

2 out of 3

- ID 5201: Robotics Seminar
- ID 6100: Robotics Laboratory
- + Six electives from the approved list

For more details refer:

https://ed.iitm.ac.in/site_resources/programs/docs/27032026_M.Tech._Robotics_Curriculum.pdf



COURSE DETAILS

Total Credits: 9

Offered in F slot, Three lectures per week, Room ED 108

Tue: 5:00PM - 5:50 PM

Wed: 11:00AM - 11:50 AM

Thu: 9:00AM - 9:50 AM

Fri: 8:00AM – 8:50 AM

Friday sessions will be used for Tutorials/Paper presentations/Seminar

GRADING

Grading (Tentative):

2 Quizzes: 10% each → Total 20%

4 Programming Assignments and 1 Tutorial: 10% each → Total 50%

Paper presentation: 10%

End Semester Exam: 15%

Classroom participation: 5%

Lecture PPT, additional notes, assignment submission will be through Moodle

Do not email assignment or reports to me or the TAs

Check Moodle regularly and ensure you are getting the announcements and other notifications related to the course

Inform in-advance if you need extensions for quizzes and assignments

ED 5315: INTRO TO FIELD & SERVICE ROBOTICS

Prerequisites:

- ID 6040: Introduction to Robotics
- Python programming

Good to have:

- Experience with robotic simulators such as Coppeliasim from Coppeliarobotics



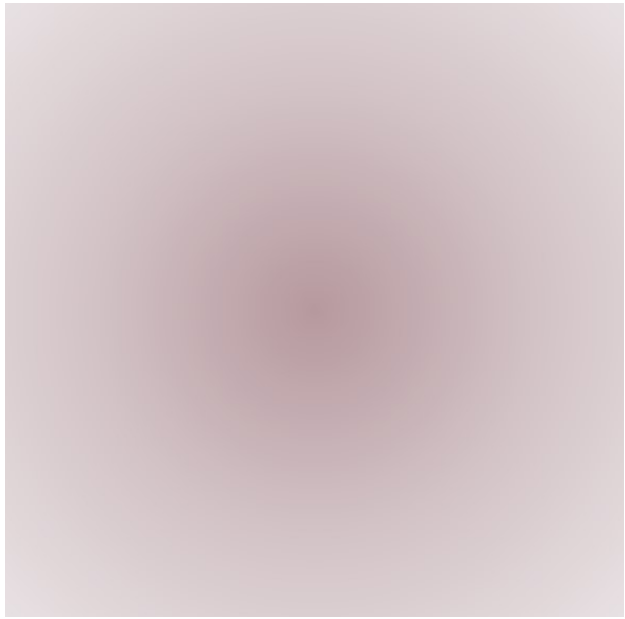
Credits: https://www.coppeliarobotics.com/

ED 5315: INTRO TO FIELD & SERVICE ROBOTICS

Objectives of the course:

By the end of the course, students would be able to:

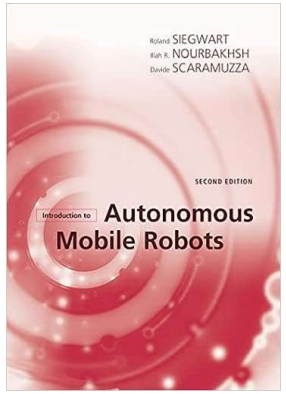
- Understand and appreciate the different challenges associated with Field and Service Robotic Systems
- Learn how to model and simulate field and service robots
- Learn the basics of control, sensing, and localization as applied to field and service robots
- Build a working autonomy stack over the course of four programming assignments and validate it over simulation

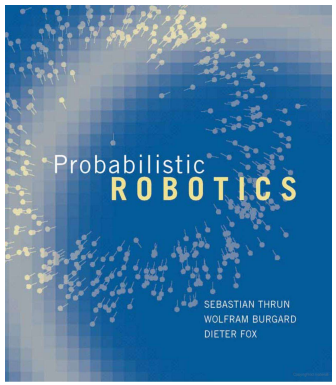


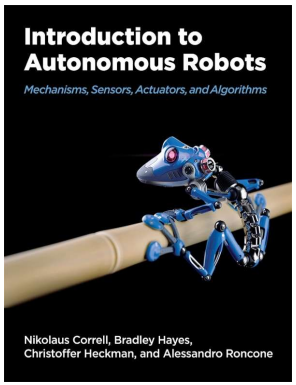
ED 5315: INTRO TO FIELD & SERVICE ROBOTICS

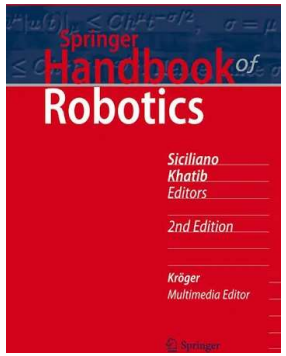
Reference books:

1. R Siegwart, I. R. Nourbakhsh, D. Scaramuza Introduction to Mobile robotics (Second Edition), The MIT Press, Cambridge
2. S. Thrun, W. Burgard and D. Fox, Probabilistic Robotics, The MIT Press, Cambridge. Available online: http://robots.stanford.edu/probabilistic-robotics/
3. Introduction to Autonomous Robots Mechanisms, Sensors, Actuators, and Algorithms. , The MIT Press, Cambridge. Available online: https://introduction-to-autonomous-robots.github.io/
4. Extra reading: B. Scicilliano, O Khatib, Handbook of Robotics, Springer





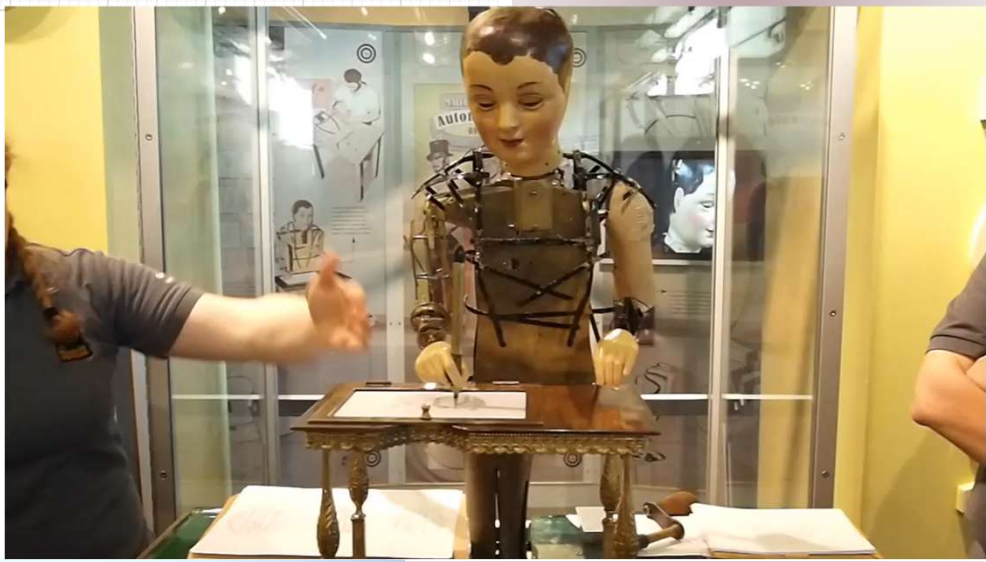


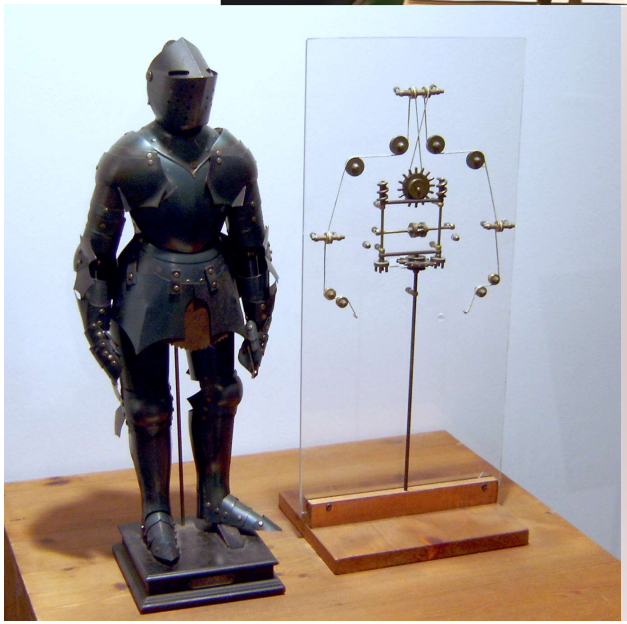


HISTORY

Automatons

The origin of robotic systems can be traced back to a class of devices that existed in the past, these systems were generally referred to as Automatons
Similar systems also existed in India





Credits:

1. Maillardet's automaton at the Franklin Institute by Z22, via Wikimedia Commons, licensed under CC BY-SA 4.0. https://creativecommons.org/licenses/by-sa/4.0/

2. Model of Leonardo's Armored Knight –1495
Photo by Erik Möller. Leonardo da Vinci. Mensch - Erfinder - Genie exhibit, Berlin 2005. https://commons.wikimedia.org/w/index.php?curid=395321

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HISTORY

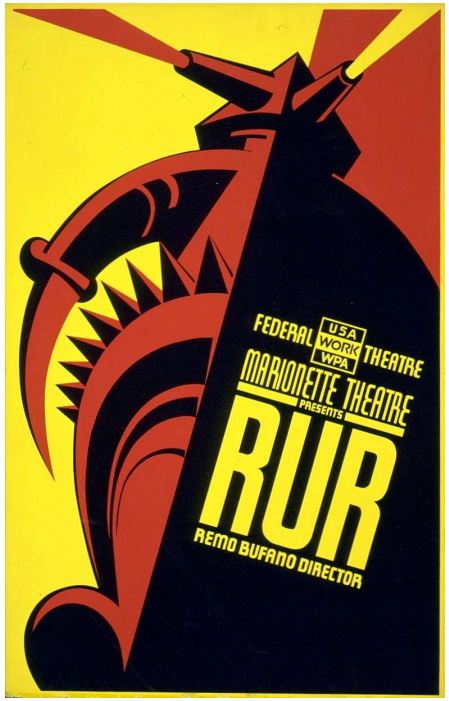
First mention of the term – Robot

- By the Czech playwright, Karel Capek, in his play titled R.U.R - Rossums Universal Robots, in the year 1920

- The word comes from the Czech term *robota*, which means tedious labor

Isaac Asimov - Runaround (1942): defines the Three Laws of Robotics

0. A robot may not injure humanity or, through inaction, allow humanity to come to harm. (This was added after the initial three laws.)
1. A robot may not injure a human being, or, through inaction, allow a human being to come to harm.
2. A robot must obey the orders given to it by human beings except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

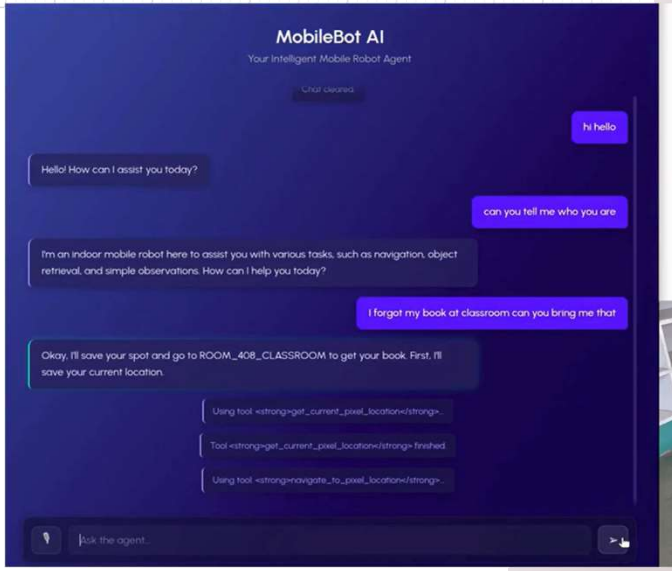


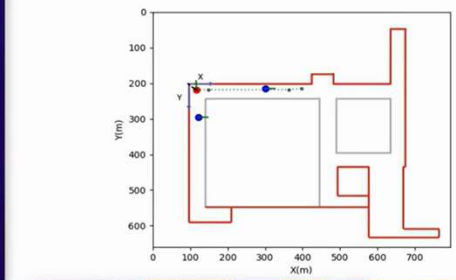


Credits: Images in public domain from Wikipedia
https://en.wikipedia.org/wiki/Karel_%C4%8Capek

TIMELINE

- 1495 Leonardo DaVinci designs a mechanical device that looks like an armored knight
- 1920 Karel Capek first introduces the word robot in his satirical drama, R.U.R (Rossum's Universal Robots)
- 1942 Isaac Asimov publishes Runaround, in which he defines the Three Laws of Robotics
- 1958 Joseph F. Engelberger and George C. Devol set up the world's first robot company, Unimation, Inc.
- 1997 NASA Pathfinder lands on Mars to explore life on the planet and send information via satellite images. Robot was powered by Solar panels
- 2000 Honda showcases Asimo, the next generation of its series of humanoid robots
- 2004-2005 DARPA Grand Challenge
- 2015 DARPA Robotics Challenge
- **Current date LLM-driven embodied-AI wave ……**

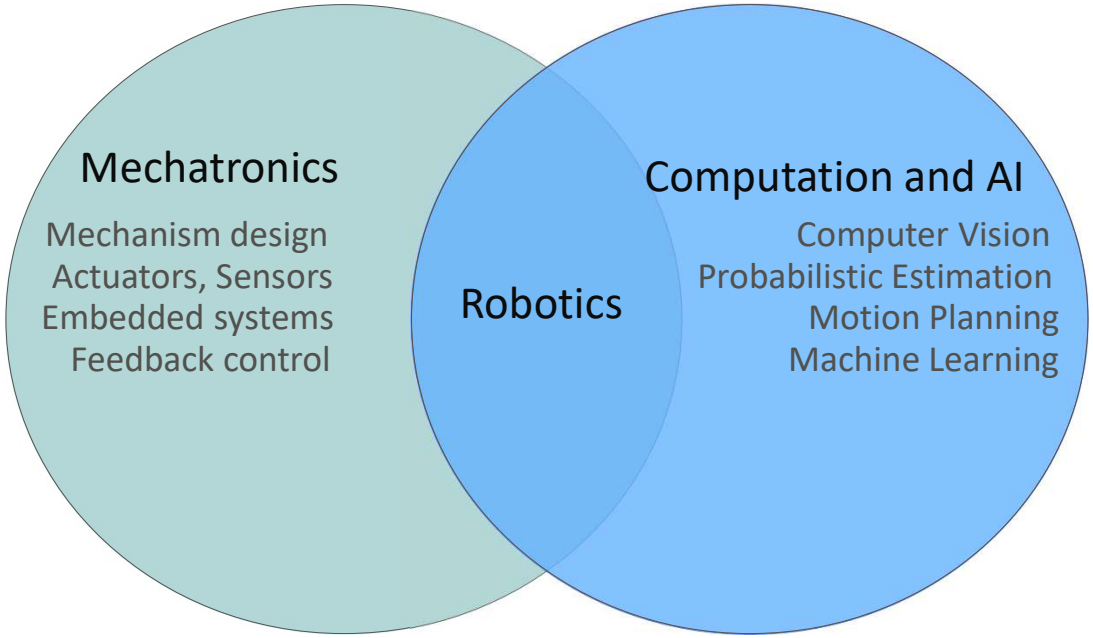






Picture credit: Autonomous Systems Lab, Dept. of Engineering Design, IIT Madras. https://asl-iitm.github.io/

DEFINITION



Mechatronics: A combination of different fields of Engineering: Mechanical, Electrical, Electronics, Computer Science

Computation and AI: A combination of different fields of computing and mathematics: algorithms, probability and statistics, computer vision, and machine learning.

Robotics draws from both domains

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DEFINITION

Robotics

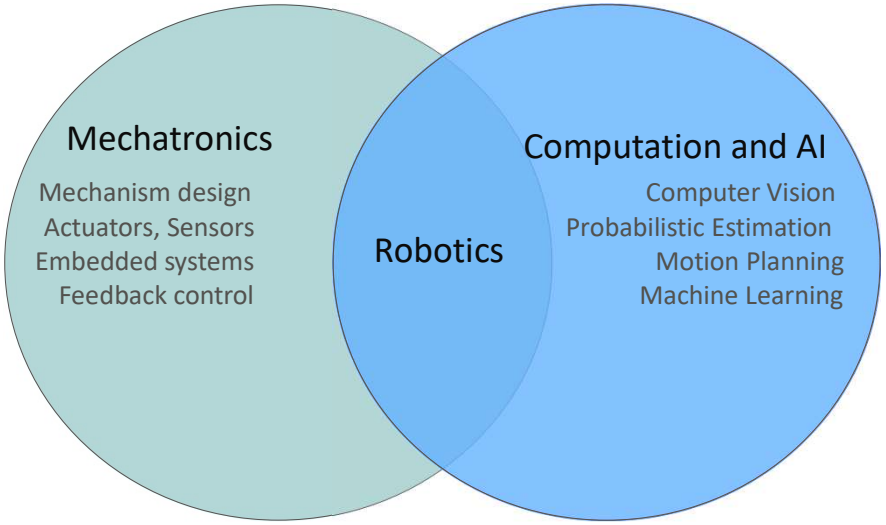
General Definitions: A hobby, a science fiction genre, a scientific/engineering discipline, or an industrial technology

Specific to this Course

Robotics: A collection of science and knowledge that is useful for design, analysis, and control of robots

Robot: Software controlled mechanical device that uses sensors to guide one or more actuators through programmed motions to perform an intelligent action

A robot is autonomous when it makes decisions in response to its environment, rather than purely replaying a pre-programmed set of motions.



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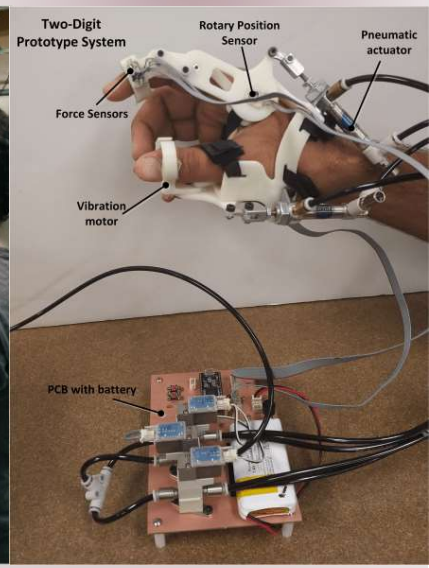
CLASSIFICATIONS

Categories of Field and Service robotic systems:

1. Intelligent grasping and manipulation systems
2. Mobile Robots: Ground robots, Surface or underwater vessels, Aerial robots, Legged robots
3. Medical robots: Surgical robots, Assistive robots
4. Multi-agent / swarm systems …..









Picture & video credit: Autonomous Systems Lab, Dept. of Engineering Design, IIT Madras. https://asl-iitm.github.io/

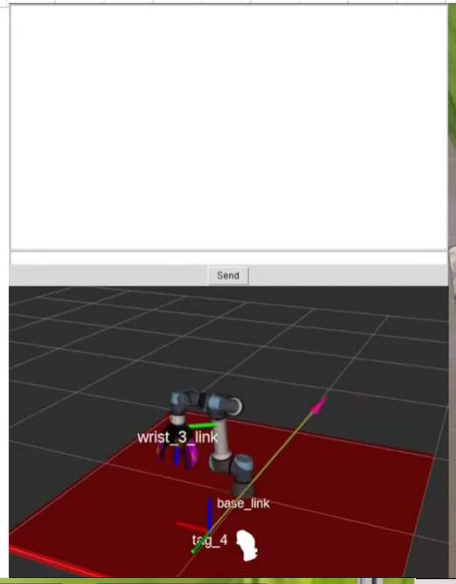
INTELLIGENT GRASPING & MANIPULATION

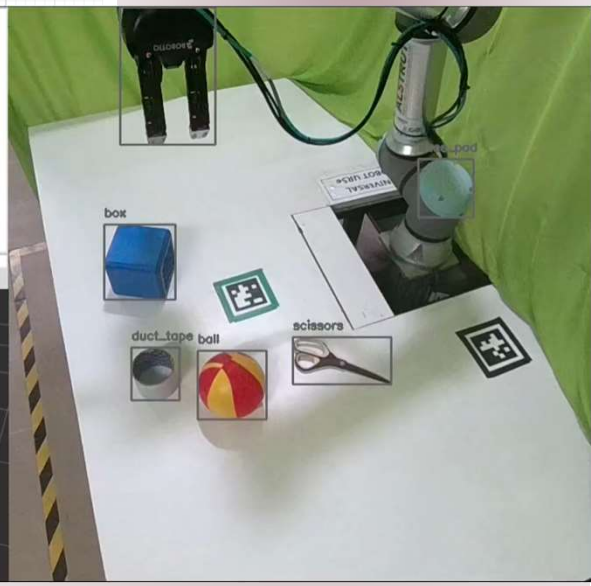
Perception - RealSense D455 RGB-D; object detection and classification in the image, Visual markers for camera-to-robot extrinsic calibration

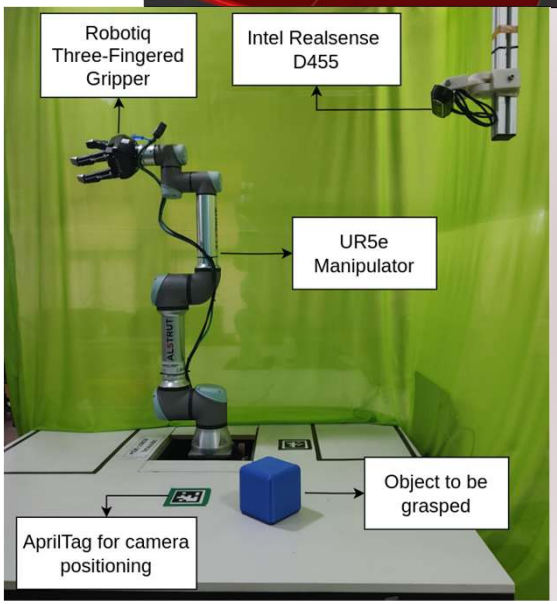
Pose estimation - Object 6-DOF pose recovered from depth “Where is the object relative to robot“

Planning - Grasp selection followed by a collision-free joint-space trajectory to the pre-grasp pose

Control - Joint-level control, Gripper control







Picture & video credit: Autonomous Systems Lab, Dept. of Engineering Design, IIT Madras. https://asl-iitm.github.io/

INTELLIGENT GRASPING & MANIPULATION

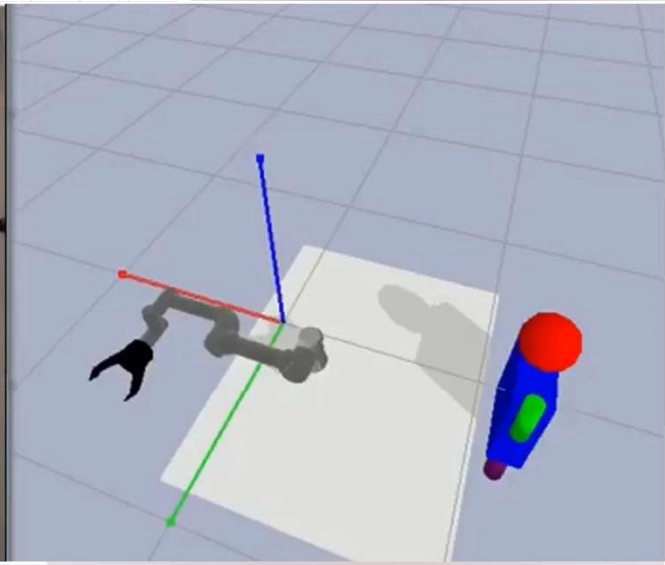
Perception - Tracking the human arm, updated every frame

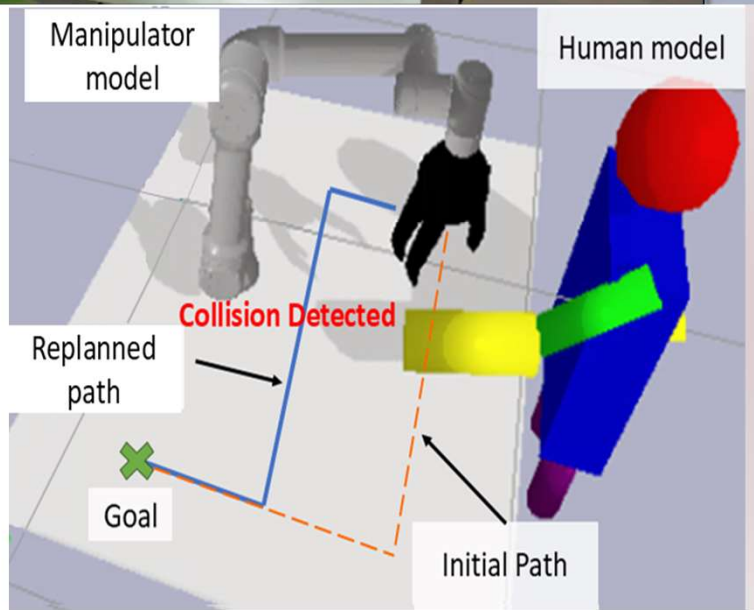
Pose estimation - Human pose estimated and tracked as a dynamic obstacle

Planning - Iterative collision prediction against human model with online re-planning

Control - Joint-level control, stop when collision is imminent, gripper control







Picture & video credit: Autonomous Systems Lab, Dept. of Engineering Design, IIT Madras. https://asl-iitm.github.io/

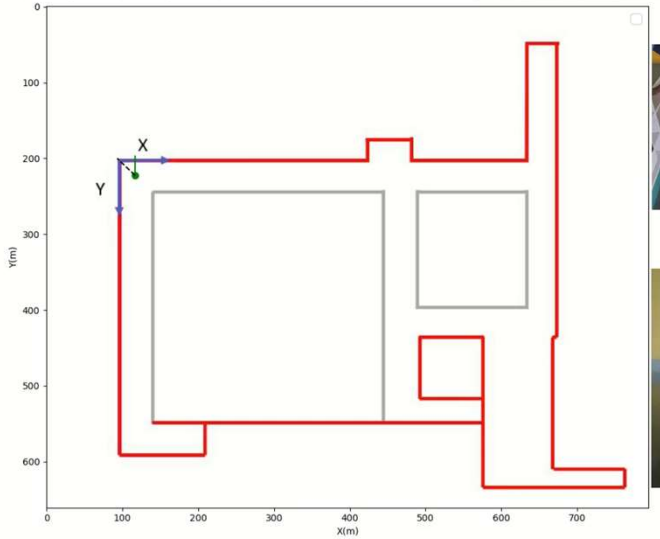
AUTONOMOUS MOBILE ROBOTS

Perception - LIDAR, camera, wheel encoders for detecting walls and other obstacles

Pose estimation - position tracking through scan-matching against the prior map

Planning - Global path over the known map, plus a local planner reacting to immediate obstacles

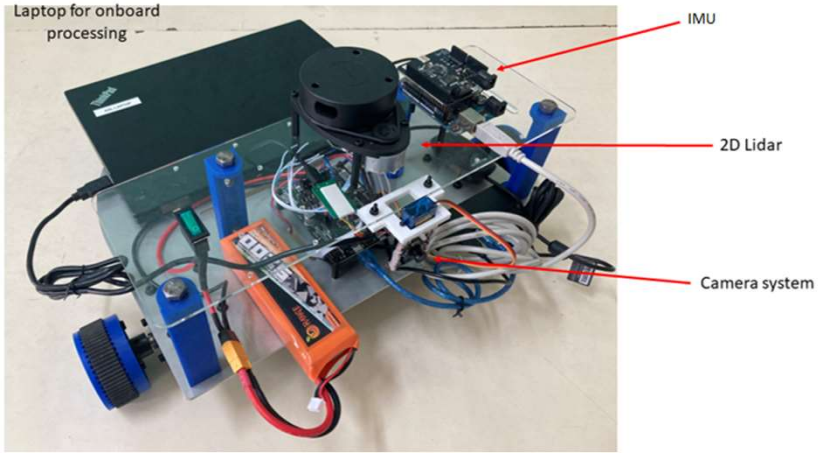
Control - Differential-drive velocity tracking the local path







Front Camera



Picture & video credit: Autonomous Systems Lab, Dept. of Engineering Design, IIT Madras. https://asl-iitm.github.io/

AUTONOMOUS MOBILE ROBOTS

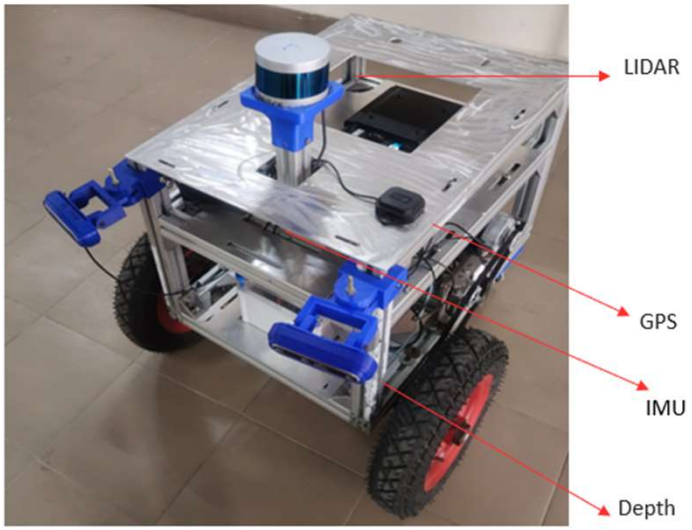
Perception - LIDAR, depth camera, IMU, GPS, encoders

Pose estimation - GPS with corrections from IMU and wheel odometry

Planning - Traversability estimation

Control - Differential-drive velocity tracking the local path





Picture & video credit: Autonomous Systems Lab, Dept. of Engineering Design, IIT Madras. https://asl-iitm.github.io/

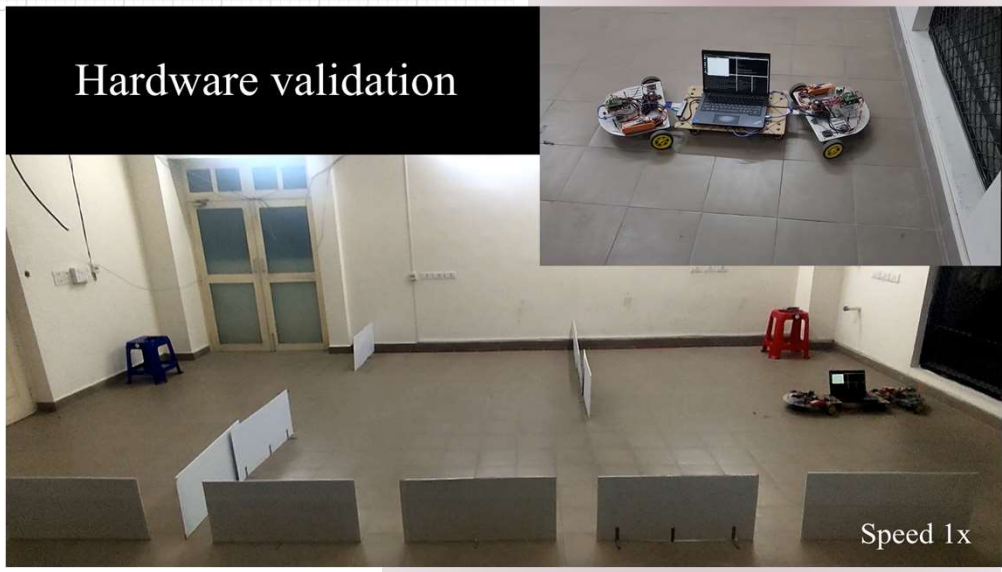
MULTI-AGENT SYSTEMS

Perception - Each robot senses its own state along with absolute pose of the payload

Pose estimation – Indoor GPS, Ultra Wide Band (UWB) system

Planning - Coordinated trajectory for the two-robot-plus-payload system

Control – Model Predictive Control with the payload kinematics and actuation constraints



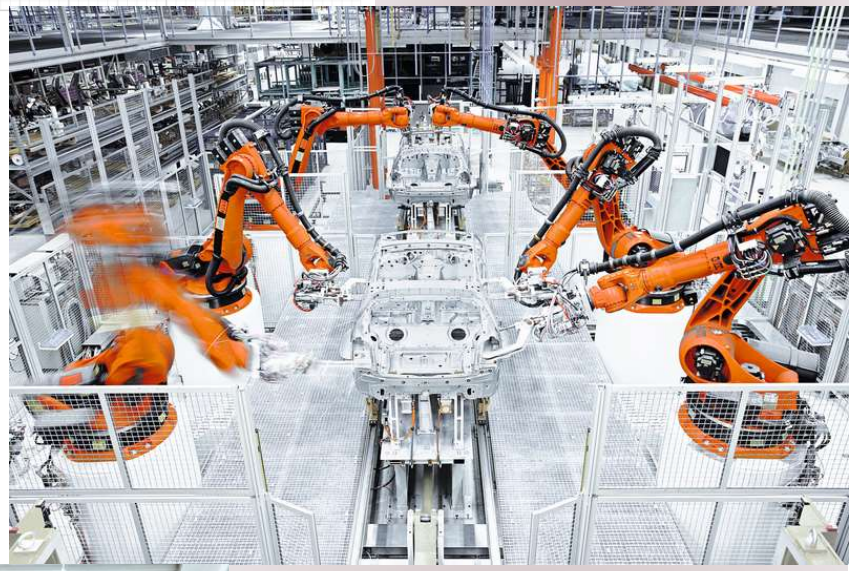


Picture & video credit: Autonomous Systems Lab, Dept. of Engineering Design, IIT Madras. https://asl-iitm.github.io/

INTRODUCTION

ID-6040 : Introduction to Robotics →

- High speed, good accuracy
- Perform repeated tasks with superhuman precision



- Lack of mobility
- Limited range of motion/application
- → Cannot be used in field

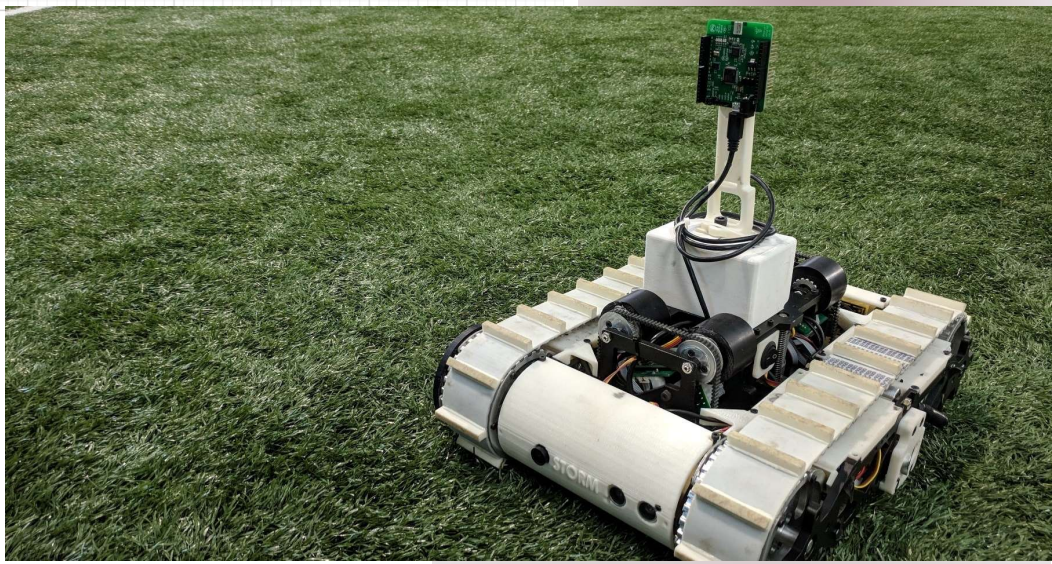


Image source: KUKA Group

INTRODUCTION

ED-5315 : Introduction to Field and Service Robotics

What are the components present in a Field Robotic System?



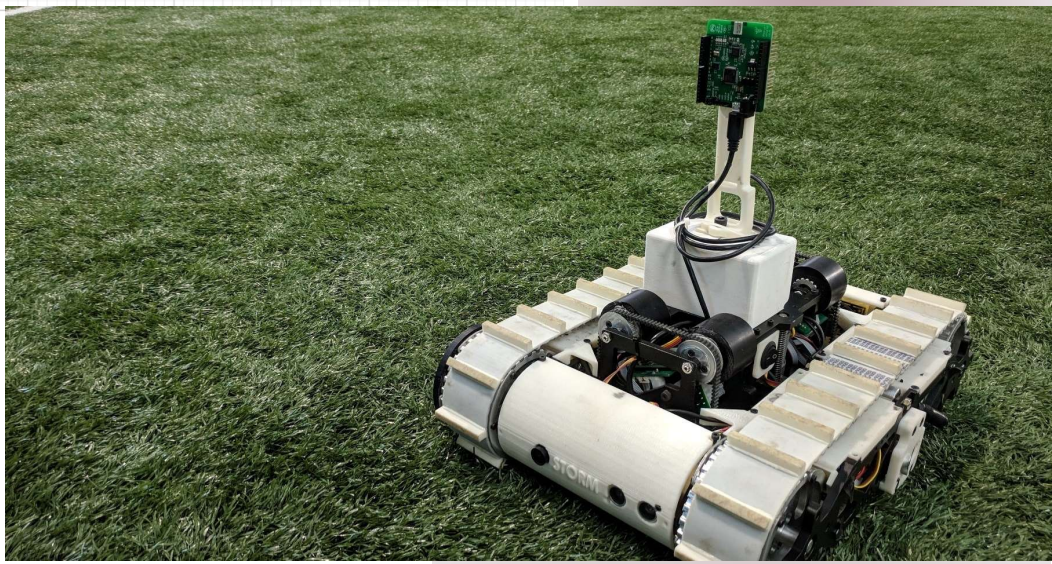
Picture & video credit: Autonomous Systems Lab, Dept. of Engineering Design, IIT Madras. https://asl-iitm.github.io/

INTRODUCTION

ED-5315 : Introduction to Field and Service Robotics

What are the components present in a Field Robotic System?

- Mechanical subsystem (arm, gripper, body and wheels, ...)
- Electrical subsystem (motors, computers, ...)
- Sensor subsystem (camera, force sensor, …)

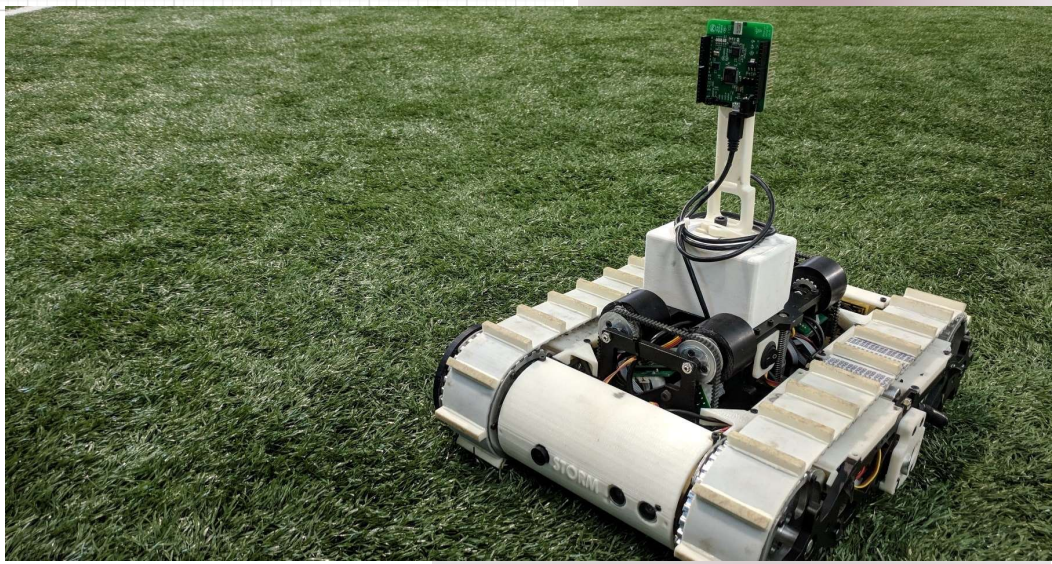


Picture & video credit: Autonomous Systems Lab, Dept. of Engineering Design, IIT Madras. https://asl-iitm.github.io/

INTRODUCTION

ED-5315 : Introduction to Field and Service Robotics

What are the capabilities needed for a Field Robotic System?

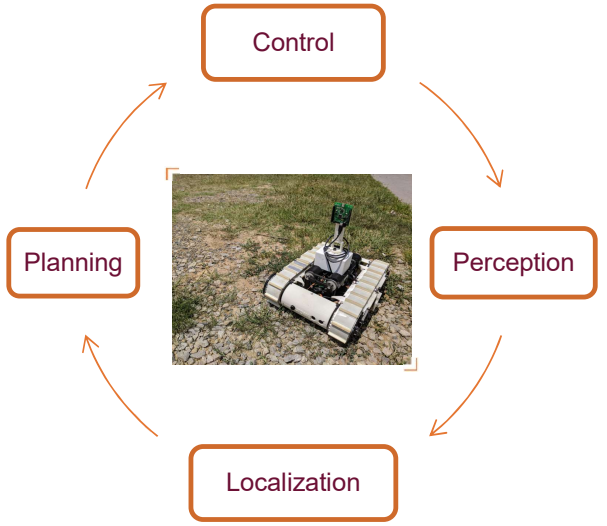


Picture & video credit: Autonomous Systems Lab, Dept. of Engineering Design, IIT Madras. https://asl-iitm.github.io/

INTRODUCTION

Capabilities needed for Field Robotic Systems:

- Perception: Sense the world - Sensor characteristics, uncertainty models
- Localization: Locate the robot on a map - Sensor fusion algorithms, Probability theory
- Planning: How to reach a goal location or accomplish a task - Artificial Intelligence
- Control: Execute the planned path - Movement/Locomotion, Kinematics, Dynamics, Actuator characteristics



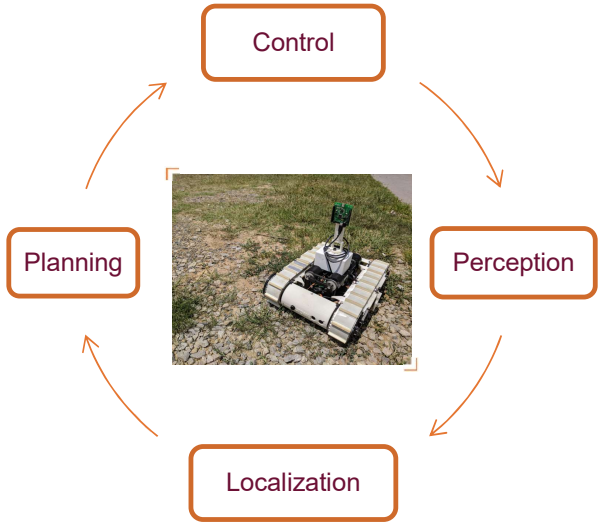
Picture & video credit: Autonomous Systems Lab, Dept. of Engineering Design, IIT Madras. https://asl-iitm.github.io/

INTRODUCTION

Building the autonomy stack through assignments:

1 Wk 5	**2** Wk 7	**3** Wk 9	**4** Wk 11
Drive	**Sense**	**Localize**	**Localize globally**
Go-to-goal control + reactive obstacle avoidance	Landmark detection + sensor-noise characterisation	Odometry	Odometry + EKF against known landmarks

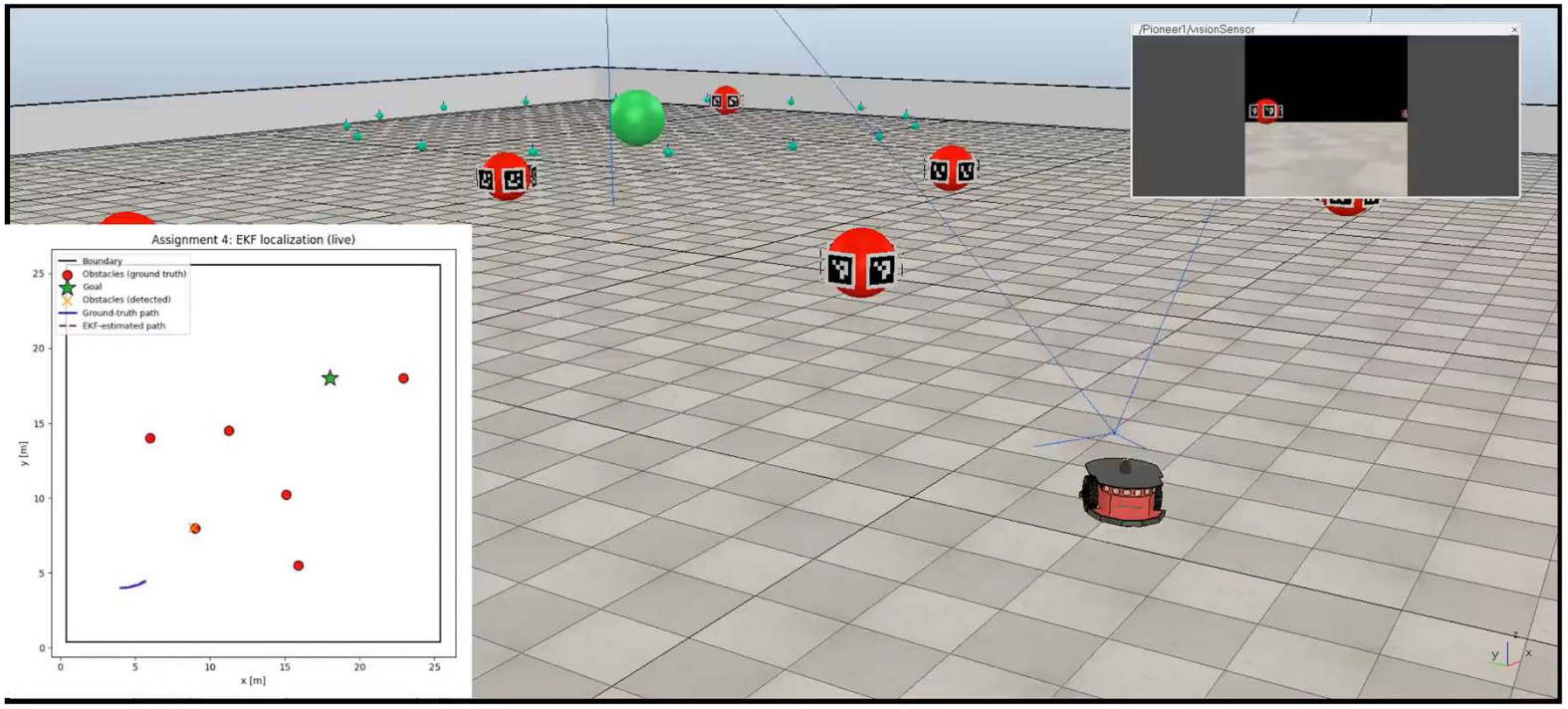
same robot, same CoppeliaSim world, one growing codebase

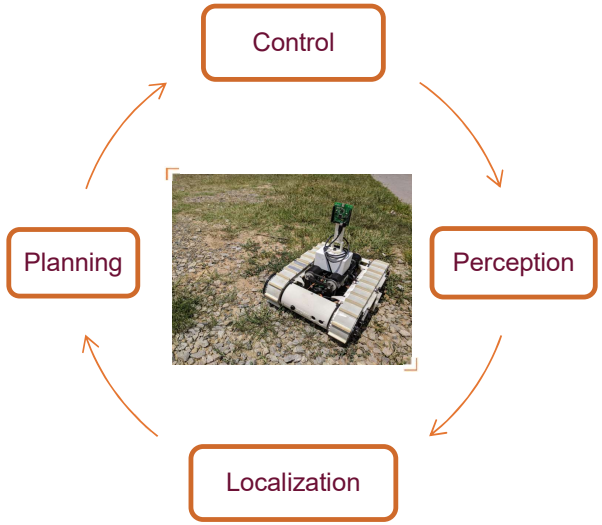


Picture & video credit: Autonomous Systems Lab, Dept. of Engineering Design, IIT Madras. https://asl-iitm.github.io/

INTRODUCTION

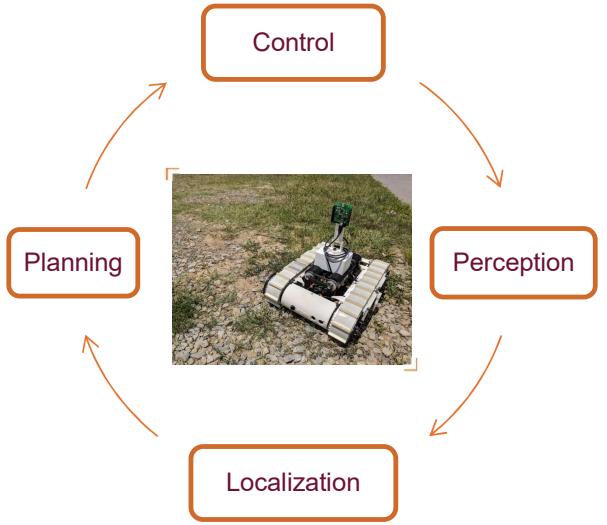
Building the autonomy stack through assignments:





Picture & video credit: Autonomous Systems Lab, Dept. of Engineering Design, IIT Madras. https://asl-iitm.github.io/

INTRODUCTION



Additional reading:

- Chapter 1 of “Introduction to Autonomous Mobile robots” book
- Chapter 1 of “Introduction to Autonomous Robots: Mechanisms, Sensors, Actuators, and Algorithms”

Picture & video credit: Autonomous Systems Lab, Dept. of Engineering Design, IIT Madras. https://asl-iitm.github.io/